RMO(INDIA) – 2002

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. In an acute angled triangle ABC, point D,E, F are located on sides BC, CA, AB respectively, such that $\frac{CD}{CE} = \frac{CA}{CB}$, $\frac{AE}{AF} = \frac{AB}{AC}$, $\frac{BF}{BD} = \frac{BC}{BA}$. Prove that AD, BE, CF are the altitudes of ABC.
2. Solve the following equation for real x: $(x^2 + x - 2)^3 + (2x^2 - x - 1)^3 = 27(x^2 - 1)^3$.
3. Let a, b, c be positive integers such that a divides b^2, b divides c^2 and c divides a^2. Prove that abc divides $(a+b+c)^7$.
4. Suppose the integers 1, 2, 3, ……, 10 are split into disjoint collections a_1, a_2, a_3, a_4, a_5 and b_1, b_2, b_3, b_4, b_5 such that $a_1 < a_2 < a_3 < a_4 < a_5$ and $b_1 > b_2 > b_3 > b_4 > b_5$.
   (a) Show that the larger number in any pair $\{a_j, b_j\}, 1 \le j \le 5$, is at least 6.
   (b) Show that $|a_1 - b_1| + |a_2 - b_2| + |a_3 - b_3| + |a_4 - b_4| + |a_5 - b_5| = 25$ for every such partitions.
5. The circumference of a circle is divided into eight arcs by a convex quadrilateral ABCD, with four arcs lying inside the quadrilateral and the remaining four lying outside it. The lengths of the arcs lying inside the quadrilateral are denoted by p, q, r, s in counterclockwise direction starting from some arc. Suppose $p + r = q + s$. Prove that ABCD is a cyclic quadrilateral.
6. For any natural number n > 1, prove that the inequality:
$$\frac{1}{2} < \frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + ... + \frac{n}{n^2+n} < \frac{1}{2} + \frac{1}{2n}.$$
7. Find all integers a, b, c, d satisfying the following relations:
   (a) $1 \le a \le b \le c \le d$.
   (b) $ab + cd = a + b + c + d + 3$.

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